Project ASTRO Mission and Scope

Assistive Systems for Trusted Robotic Operations

Revision R0.1 | 2026-10-23 | Owner Faisal Farouq

Task control

Artifact: ASTRO-A-01

Objective: Define the controlled first-year caregiver-support research mission.

Required inputs: Owner brief SRC-01; fallback appearance description; the five owner decisions.

Known values and provisional assumptions: 1.829 m target, <=55 kg target, >65 kg major review, 10 lower-body joints, USD 5000-10000 funding range; no concept image available.

Unknowns: Funding, facility, reviewers, equipment access, selected actuators and approved recipient.

Calculations: Three-step distance example: 3 x 0.15 m = 0.45 m. At 0.10-0.15 m/s this interval takes 4.5-3.0 s; stride and timing are provisional.

Output file: requirements/01_mission_and_scope.md

Implementation steps: 1. Review scope and exclusions. 2. Record owner dispositions D-01 to D-05. 3. Baseline accepted requirements at G0. 4. Begin software-only work.

Acceptance test: One-page mission contains all measurable year-one outcomes and exclusions; owner acceptance is separately recorded.

Stop conditions: Conflicting scope, real patient data, unapproved spending or motion.

Result: PASS - document coverage only; owner acceptance pending

Next dependency: G0 owner dispositions

Physical authorization and supervision: No physical work authorized.

Budget effect: USD 0 committed; cap unchanged.

Revision: R0.1

Date: 2026-10-23

Accountable owner: Faisal Farouq

Specification

The Project ASTRO Caregiver Support Robot will help caregivers with companionship, low-risk engagement, fictional record organization and communication. Faisal Farouq remains the accountable project owner. Caregivers retain authority, responsibility and clinical judgment. The robot will identify itself as a research support system and will never claim professional-care or emergency-response authority.

Year 1 is a controlled research demonstration. The target robot is 1.829 m tall and no more than 55 kg in operating configuration. A projection above 55 kg requires rework; above 65 kg requires a major design review. The lower body has 10 active joints. The goal is three consecutive slow tethered steps in each of two separate trials, targeting 0.10-0.15 m/s inside a cleared controlled test area with rated robot fall protection. The financial ceiling is USD 10000; feasibility at either USD 5000 or USD 10000 remains unproven.

The companion experience will be developed first on a laptop using fictional profiles, simulated observations, games, medication-response logging, audit history, approved-recipient summaries and telepresence handoff. Every simulated reading retains its

value, units, source and time. Reports require exact-content preview and explicit send confirmation. Only approved test recipients may receive them. Endpoint receipt never implies human review.

A separate handling demonstration will move a 2 kg inert object through an approved controlled trajectory for 10 supervised repetitions while the robot is stationary and mechanically secured. Walking with the test object is excluded. The handling module is included in onboard mass accounting; the test object is additional only in the stationary configuration.

Appearance follows the written fallback: removable rigid white rounded plastic panels, visible narrow seams, charcoal protected joint gaps, subtle teal, simple connected eyes and broad flat rounded-rectangle feet. Motion clearance, cooling, service access, sensing, stop access and arrest attachments take priority over cosmetic proportions.

Excluded are real patient data, live EHR integration, diagnosis, medical image interpretation, treatment or dose recommendations, dispensing or medication administration, person lifting/support/restraint, unsupervised high-energy motion and unapproved public operation. A 5 kg payload, untethered gait and clinical use are later-phase questions. No physical performance, fabrication, approval or clinical validation is claimed by this baseline.